HOW HUMANS LEARN UNFAMILIAR WORLDS: A THINK-ALOUD DATASET ON ARC-AGI-3

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ABSTRACT

Learning through exploration and using newly acquired knowledge to achieve goals are central capabilities of general intelligence. Understanding these capabilities requires tracing how learners build, use, and revise world models during interaction. We construct the first dataset that records how humans discover and use the mechanisms of novel environments, using ARC-AGI-3 as a testbed: 144 trajectories from 64 participants across 25 public games, each pairing time-synchronized actions with think-aloud audio, plus 250 trajectories from large language model (LLM) agents with and without harnesses on the same games. We annotate mechanism triggers and recognition, action purposes, and errors and recovery, linking changes in understanding to the observations and actions that support them. We analyze characteristics of human cognition and compare them with those of LLM agents. Both groups anticipate mechanisms and use familiar analogies. Qualitative representations are more common in successful human trajectories than in successful LLM trajectories (50.0% versus 14.3%). Humans more often explore unfamiliar situations without a concrete hypothesis, while hypothesis testing is common in both groups. We also trace how learners recover through belief revision, replanning, and execution correction. As LLMs with well-designed harnesses reach saturated performance on public ARC-AGI-3 games, our cognitive analyses motivate next-generation benchmarks that challenge familiar priors, require accumulated evidence and abstraction, and test planning through subgoals, experimentation, and recovery.

1 INTRODUCTION

Discovering and using unfamiliar rules is central to general intelligence (Legg & Hutter, 2007; Chollet, 2019). Actions both reveal and change an environment, requiring learners to gather information while pursuing goals. Unexpected outcomes can prompt experiments, revised rules, or new plans. Humans learn new video games within minutes (Tsividis et al., 2017; Lake et al., 2017), solve physical puzzles in few attempts (Allen et al., 2020), and experiment with unfamiliar toys even as children (Cook et al., 2011; Schulz, 2012). Interactive benchmarks compare agents with humans on success and action efficiency (ARC Prize Foundation, 2026a; Warrier et al., 2026), with large language model (LLM) agents rapidly closing the gap (Han et al., 2026; Zeng et al., 2026). These scores leave cognitive processes largely unobserved. Studying them requires tracing what learners understand as they explore and act.

Existing human data rarely reveal what learners understand at each step. Action traces alone are ambiguous: repeatedly pushing an object could reflect experimentation, executing a solution, or

persisting under an incorrect belief. ARC-AGI-1 studies (Chollet, 2019) record solutions, editing traces, rule descriptions, and instructions (Johnson et al., 2021; LeGris et al., 2025; Acquaviva et al., 2022; Kim et al., 2026), but concern generalizing from examples rather than exploring an environment. The ARC-AGI-3 human release records actions but not their underlying reasoning (Kamradt, 2026). Think-aloud datasets pair reasoning with behavior (Wurgaft et al., 2025; Liu et al., 2026), but do not require discovering both unfamiliar dynamics and goals. Studying this process requires concurrent reports aligned with actions and environmental responses.

We construct the first dataset tracing how humans discover and use the *mechanisms* of novel interactive environments (Figure 1). Our testbed, ARC-AGI-3, requires players to infer mechanisms and goals through interaction without instructions (ARC Prize Foundation, 2026a). Its games rely on core object and spatial priors; successive levels introduce or recombine mechanisms. Using concurrent think-aloud (Ericsson & Simon, 1993), we collect 144 successful trajectories from 64 participants across 25 public games, with 136,876 actions aligned with speech, including errors and recovery. We also collect 250 public LLM trajectories on these games: 100 from direct models and 150 using VISTA, Schema, OPINE-World, and Astra harnesses (Han et al., 2026; Schema contributors, 2026; OPINE-World contributors, 2026; ARC Prize Foundation, 2026c).

Our annotation framework covers three components: world-model changes, action generation, and errors and recovery (Section 3). We catalog 354 game mechanisms, locate their triggers and recognition, classify the purposes of action segments, and link errors to subsequent evidence of correction. This alignment supports examining how understanding develops and guides action throughout a task.

Section 4 examines prior knowledge, qualitative representations, action purposes, and errors and recovery. Both humans and LLMs anticipate mechanisms and use familiar analogies. Qualitative descriptions occur in 50.0% of 144 successful human trajectories versus 14.3% of 119 successful VISTA, Schema, and OPINE-World trajectories. Both groups act to pursue goals, gather information, or both. Humans more often explore without a concrete hypothesis; hypothesis testing is common in both groups. Recovery involves belief revision, replanning, and execution correction: 11 of 15 incorrect mechanism conclusions in human reports and 451 of 593 in successful LLM records are subsequently corrected. As well-designed harnesses saturate performance on public ARC-AGI-3 games, these analyses motivate benchmarks that challenge familiar priors, require accumulated evidence and abstraction, and test planning through subgoals, experimentation, and recovery (Section 5).

Our main contributions are as follows:

- **Dataset.** We release the first dataset of concurrent human think-aloud reports aligned with actions and environmental feedback in environments whose mechanisms and goals learners must discover, together with matched LLM agent trajectories and mechanism-level annotations of triggers, recognition, action purposes, errors, and corrections.
- **Analysis of humans and LLM agents.** We use aligned records to examine prior knowledge, qualitative world-model representations, action purposes, and errors and recovery, comparing human and LLM evidence and identifying questions for further study.
- **Implications for benchmark design.** We propose benchmarks with mechanisms that defy familiar priors, require cumulative induction, and extend planning through subgoals and recovery.

2 RELATED WORK

We review two lines of work: (1) datasets of human behavior and reasoning, which lack concurrent reports from people exploring unfamiliar environments; and (2) ARC-AGI-3 and its agent harnesses, which provide a setting for collecting such reports and public LLM trajectories to compare them against.

Human behavioral and reasoning-process datasets Think-aloud protocols provide evidence about reasoning during problem solving (Ericsson & Simon, 1993). Recent work scales their transcription and annotation in the Game of 24 (Wurgaft et al., 2025), uses verbal reports to constrain

cognitive models of risky choice (Xie et al., 2026), and compares human and LLM-generated reasoning in chemistry tutoring (Borchers et al., 2026). EgoTL aligns spoken reasoning with egocentric video and actions in household tasks (Liu et al., 2026).

ARC-AGI-1 tasks require learners to infer grid transformations from examples and apply them. Studies including H-ARC collect human solutions, action traces, and rule descriptions (Johnson et al., 2021; LeGris et al., 2025); LARC captures language used to communicate solutions (Acquaviva et al., 2022), and ARCTraj records object-level editing trajectories (Kim et al., 2026).

These resources cover mathematical problem solving, choices between specified options, chemistry exercises, everyday household activities, and abstract grid transformations. They do not provide concurrent reports of humans discovering both the dynamics and goals of unfamiliar interactive environments. Our dataset fills this gap by recording thoughts as people explore such environments, aligning their evolving hypotheses and plans with the actions and feedback that inform them.

ARC-AGI-3 and agent harnesses ARC-AGI-3 provides a suitable setting for studying how people explore and learn an unfamiliar environment: players must discover its mechanisms and goals through interaction (ARC Prize Foundation, 2026a;b). Building on ARC's emphasis on skill acquisition (Chollet, 2019), it presents colored grids that players interact with through key presses or clicks. Players infer the mechanisms and completion conditions from visual feedback without natural-language task instructions. The games build on basic priors such as objects and space and require no specialized knowledge. Each game comprises multiple levels, which may introduce new mechanisms or require combining known mechanisms in a new way. Progress therefore involves discovering relevant mechanisms and applying them to reach each level's completion condition; the benchmark measures performance relative to human action efficiency (ARC Prize Foundation, 2026a). We refer to individual games by their identifiers, such as BP35 and TU93.

The official ARC-AGI-3 human release provides gameplay replays in environments with unknown dynamics (Kamradt, 2026). These replays record actions and outcomes but do not include concurrent verbal reports of the hypotheses, expectations, and plans behind them. We collect these reports to study how people discover and use the environment's mechanisms.

Recent work designs harnesses that achieve strong performance on the public ARC-AGI-3 environments (Han et al., 2026; Zeng et al., 2026). Their designs support memory, model construction, and planning: VISTA stores retrievable visual experience (Han et al., 2026), Schema constructs executable models (Zeng et al., 2026), and OPINE-World combines hypothesis testing, model synthesis, and replay verification (Courtis et al., 2026). We collect the LLM trajectories and reasoning records that have been released publicly, including those from VISTA, Schema, and OPINE-World, to compare how LLMs and humans form world models, choose actions, and correct errors during interaction (Han et al., 2026; Schema contributors, 2026; OPINE-World contributors, 2026).

3 METHOD

In this section, we describe our method for dataset construction towards the goal of exploring the process of dynamically learning world models in humans and agents. We introduce the preliminaries on world model learning in Section 3.1, describe data collection in Section 3.2, and present the data annotation process in Section 3.3.

3.1 PRELIMINARIES

Interactive learning in unfamiliar environments is a central component of general intelligence (Chollet, 2019; Ying et al., 2025). Throughout interaction, learners build and revise a *world model*: their understanding of the environment's objects, relations, dynamics, and goals. This understanding guides the actions they take, whose outcomes provide further evidence for updating it (Warrier et al., 2026). We describe this recurring process as

$$(a_t, W_{t+1}) = \mathrm{Think}(W_t, \mathcal{T}_{\leq t}), \tag{1}$$

where W_t is the current world model, $\mathcal{T}_{\leq t}$ contains the observations and action–outcome history available before action a_t, and W_{t+1} is the revised model. The action produces a new observation that enters the next round of thinking.

To analyze these cognitive processes, following Zhou et al. (2025), we represent a world model as a list of mechanisms $W = (m_1, \ldots, m_K)$. Each mechanism is a conditional state-update rule:

$$m: \quad C_m(x_t, a_t) = 1 \implies x_{t+1}[V_m] = E_m(x_t, a_t).$$

Here, x_t is the full game state and a_t is an action. The condition C_m returns 1 when the mechanism applies and 0 otherwise. The effect function E_m outputs the new values of the state variables V_m affected by the mechanism; $x_{t+1}[V_m]$ denotes those variables in the next state. A mechanism specifies a local update; multiple mechanisms jointly determine the transition. Table 1 gives natural-language examples of mechanisms from the game BP35.

Table 1: Mechanisms from BP35.

```
M1 if   left pressed and left cell
         clear
   then player moves one cell left
M2 if   left pressed and solid
         block left
   then player's x-position is
         unchanged
M3 if   gravity up and cell above
         clear
   then player moves up
M4 if   green support clicked
   then support is removed
```

3.2 Data collection

We use the concurrent think-aloud method (Ericsson & Simon, 1993) to collect people's actions and their thoughts during gameplay. We recruit paid participants through Prolific. Before the formal tasks, participants read the study instructions, provide informed consent, complete a microphone check, and receive a tutorial on the interface controls. They then complete an attention-check level while thinking aloud; passing this check is required to enter the formal study. Each participant is asked to complete three games. Participants explore the unknown mechanics and goals while continuously verbalizing their thoughts as they attempt to win. Instructions emphasize detailed verbalization over rapid completion, and an on-screen reminder prompts participants to continue speaking after a period of silence. We record game actions and audio concurrently, preserving behavior and verbal reports during exploration, mechanism learning, and plan execution. Appendix A provides the instructions, time limits, and further protocol details.

The human corpus comprises 144 successful trajectories from 64 participants across 25 public ARC-AGI-3 games, containing 136,876 environment actions with concurrent verbal reports. Figure 1 shows three collected human trajectories from the first levels of LS20, BP35, and FT09, including the participants' attempts, revisions, and eventual solutions. Each trajectory pairs gameplay frames with concurrent think-aloud excerpts by action step. These paired records support cognitive analyses of how people understand unfamiliar environments, choose actions, and solve interactive tasks.

We also collect publicly released action trajectories and available reasoning records from different LLMs, both in direct configurations and with harnesses including VISTA, Schema, OPINE-World, and the Astra Provider Adapter, to compare human and LLM exploration, world-model induction, and planning processes (Han et al., 2026; Schema contributors, 2026; OPINE-World contributors, 2026; ARC Prize Foundation, 2026c). This corpus contains 250 trajectories across the same 25 games: 100 from direct agents and 150 from agents with harnesses, with one trajectory per game for each configuration. System configurations and comparison scope are detailed in Appendix D.

3.3 Data annotation

We annotate three aspects of human and LLM interaction: the cognitive processes involved in forming a world model, the purposes of actions, and errors and recovery. Human think-aloud reports and public agent records are aligned with game actions and observations, so that each annotation can be examined in its interaction context (Figure 2).

3.3.1 Changes in the world model

Game mechanism annotation. We manually annotate the mechanisms designed into each ARC-AGI-3 game. Two authors jointly annotate the mechanisms through discussion, describing each mechanism in natural language. Some mechanisms are not required for game completion, such as failure conditions and boundary conditions on the agent's maximum movement range. Across 25 games, the catalog contains 320 mechanisms required for completion and 34 optional game mechanisms.

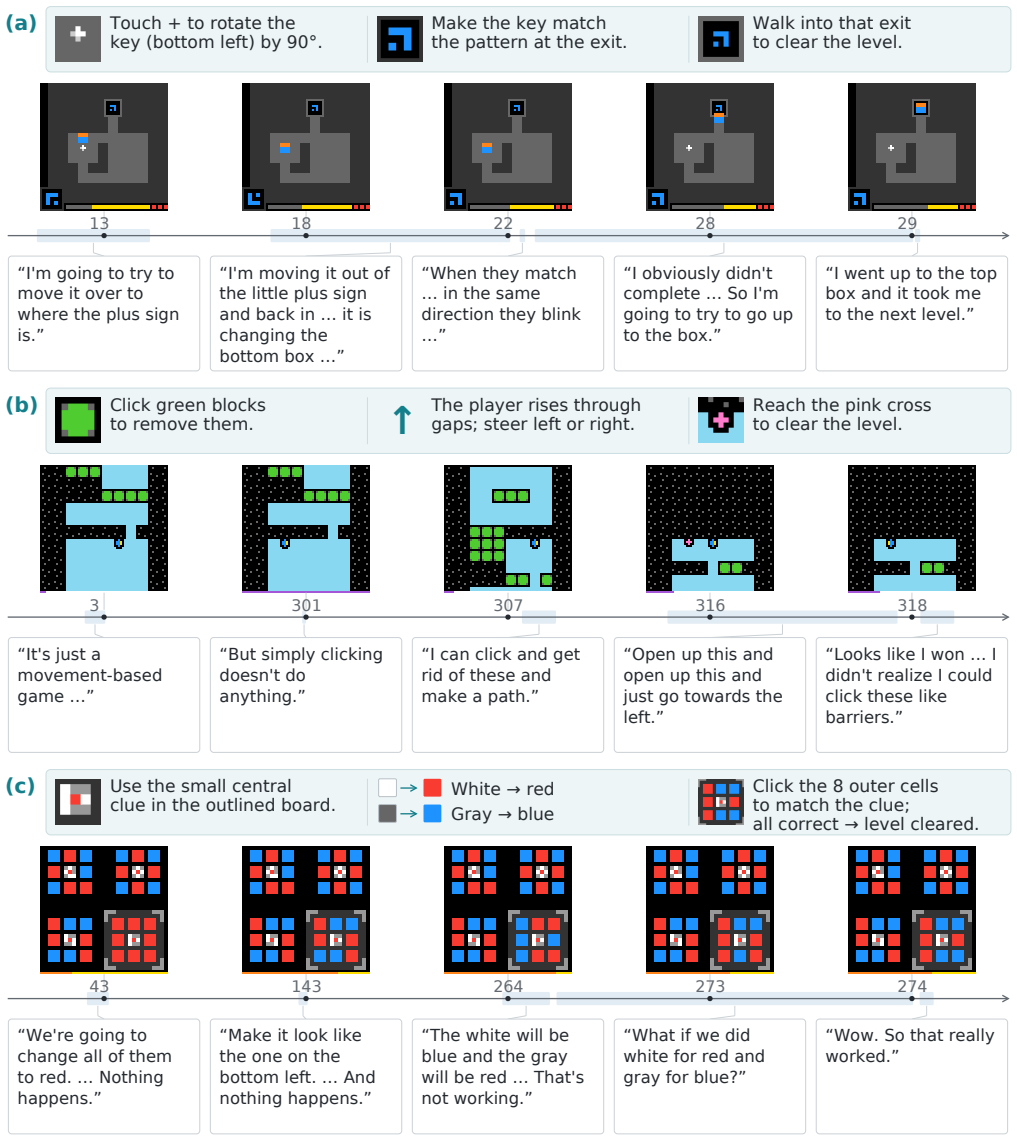


Figure 1: **Human trajectories and concurrent think-aloud data.** Key frames and aligned think-aloud excerpts trace participants' attempts to solve the first level of (a) LS20, (b) BP35, and (c) FT09. Each row includes the key mechanics needed to solve that level.

Mechanism trigger annotation. For each trajectory, we annotate when each mechanism is triggered: the times at which its conditions are satisfied and it affects the next state. We insert trigger-detection code into the ARC-AGI-3 game source. The detectors check the values and changes of mechanism-specific variables before and after actions, as well as whether relevant code paths execute, to locate mechanism triggers. We use GPT-5.6 Sol to implement the detectors, and one author spot-checks the resulting annotations. All 30 manually checked trigger annotations were correct.

Cognitive changes in the world model. We infer changes in people's world models from their trajectories and think-aloud reports. Participants sometimes describe changes in their understanding in detail; at other times, they express an insight only through an interjection, in which case we infer their understanding indirectly from subsequent actions. We provide GPT-5.6 Sol with timestamped transcripts and trajectory data and ask it to infer changes in the world model wherever the available evidence permits. Because participants do not verbalize every change in their thinking, their

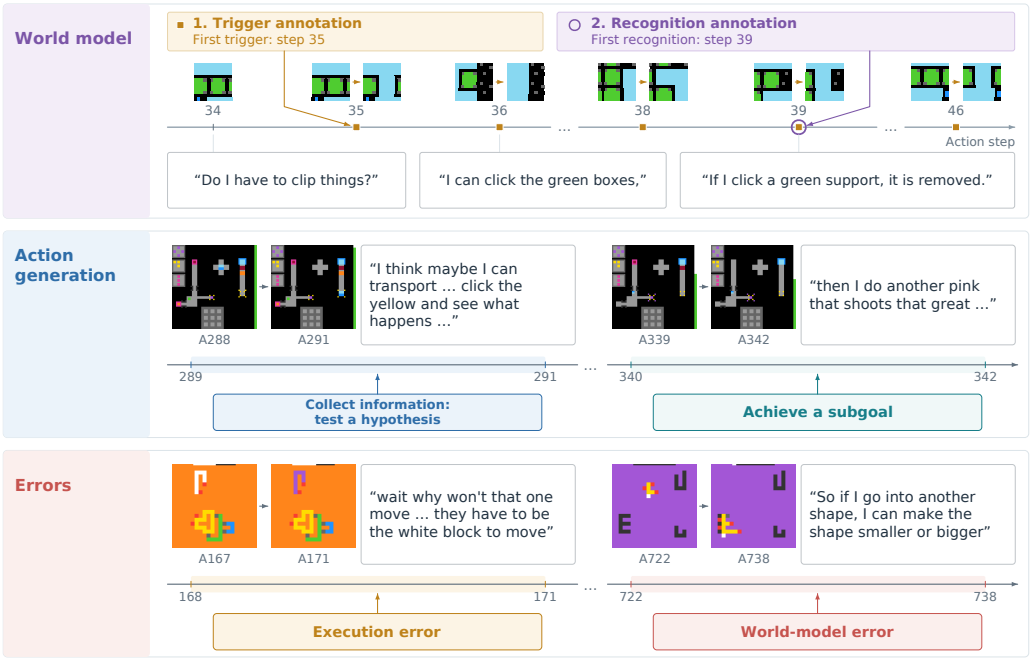


Figure 2: **Data annotation.** Annotations cover three components: *world-model changes*, tracking mechanism triggers and when and how learners recognize mechanisms; *action generation*, distinguishing goal pursuit, information gathering, and actions serving both purposes; and *errors and recovery*, identifying mechanism, planning, and execution errors and linking them to any subsequent corrections.

understanding of some mechanisms remains unclear. In particular, we annotate the first time each mechanism is correctly recognized and the participant's description of that mechanism. Recognition can precede or follow the first trigger. We manually reviewed 100 recognition annotations; 97 were correct (97% accuracy).

3.3.2 Action generation

We annotate the purposes of contiguous action segments using the associated verbal or reasoning records and the surrounding observations. GPT-5.6 Sol classifies action purposes, and we manually review the category assignments. The principal categories are achieving a goal or subgoal, collecting information to update the world model, and serving both purposes. Information-seeking purposes further divide into exploring an unfamiliar situation without an existing hypothesis, testing a proposed mechanism, and investigating an unexpected outcome. One author independently reviewed 30 action segments; all 30 classifications were correct (100% accuracy).

3.3.3 Errors and recovery

We annotate errors throughout both human and LLM trajectories, distinguishing incorrect mechanism beliefs, planning errors, and execution errors. A mechanism error is an accepted belief that conflicts with the game's actual behavior; a tentative hypothesis alone does not establish such an error. Planning failures are attempts that do not achieve an explicit goal or subgoal, while execution errors occur when the actual input deviates from the intended operation. We use the game source, observed transitions, and the learner's statements and actions to establish the mistake and locate the supporting interval. We also record explicit expressions of not understanding a rule without a concrete explanation, and uncertainty about the level goal, which can coexist with candidate goals. Continuous expressions concerning the same issue are merged, and these unknowns are distinguished from accepted but incorrect conclusions.

We then link each error to any later evidence of correction. Correcting a mechanism error requires replacing the same incorrect belief; correcting a planning or execution error requires addressing the mistaken plan or action. We retain the evidence for both the error and its correction, including cases where correction is not observed. Section 4.4 reports event counts and the numbers of observed corrections or subsequent understanding.

4 DATA ANALYSIS

Our corpus preserves state changes, actions, and concurrent thought reports during interactive problem solving, providing rich primary material for studying how humans understand unfamiliar environments. The following analyses examine selected cognitive phenomena and compare human trajectories with those of LLMs to identify similarities and differences. These analyses only begin to explore the corpus's research potential. Researchers can return to the original records to develop and test further hypotheses about cognition in interactive learning.

4.1 HUMANS DRAW ON PRIOR KNOWLEDGE

We find that humans can infer the existence of a mechanism before it is first triggered. Comparing each mechanism's first trigger with its first correct recognition, we find that 57 of 1,425 paired mechanism instances (4.0%) are recognized before their first trigger. We interpret this as evidence that prior knowledge helps people anticipate how an unfamiliar environment works. For example, a participant in TU93 identifies the green endpoint as the goal and moves toward it before reaching it: recognition is recorded before action 14, while the first trigger occurs after action 23 (Figure 3).

LLMs also exhibit this property. The proportions of mechanisms recognized before their first trigger are 11.3% and 9.6% for VISTA–Claude and VISTA–GPT, 10.8% and 9.2% for Schema–Claude and Schema–GPT, and 7.3% for OPINE–Claude (Table 3). Thus, anticipatory recognition is present in both groups and is more frequent in these LLM configurations.

People also use analogies to familiar mechanisms to form expectations about an unfamiliar world model. Analogies to knowledge outside the current game occur in 37/144 human trajectories (25.7%). For example, after an attempted interaction, a BP35 participant reports having interpreted an object as a Mario-style teleport pipe, then rejects that interpretation based on the feedback (Figure 4). Such analogies supply an initial account that can be refined through interaction.

LLMs likewise use analogies, in 26/119 successful trajectories (21.8%): a VISTA agent relates LF52 to peg solitaire, while a Schema agent interprets FT09 through Lights Out. Familiar structures therefore guide world-model construction in both humans and LLMs.

4.2 HUMANS USE QUALITATIVE REPRESENTATIONS

Humans often form a coarse understanding of a mechanism that leaves some of its details unspecified. We refer to this as a *qualitative representation*: the learner expresses how a mechanism works without specifying all relevant conditions, timing, or magnitudes. For example, a TU93 participant describes an enemy as following the player's path and plans to lead it on a detour, without specifying its movement delay. In G50T, a participant initially describes a gray clone as following them, without specifying how its replay is coupled to successful player moves. These examples show how a coarse account of a mechanism can guide further reasoning and action.

Qualitative descriptions occur in 72 of the 144 human trajectories (50.0%). LLMs also use qualitative representations, in 17 of the 119 successful trajectories (14.3%). Qualitative descriptions are therefore more common in human reports. LLM records also contain precise numerical descriptions. For example, Schema–GPT explicitly records a six-pixel downward movement in G50T; OPINE–Claude represents a patrolling enemy's route in TU93 by five row coordinates, $[16, 22, 28, 34, 40]$, and identifies a round-trip period of eight valid moves.

4.3 ACTION PURPOSES

We analyze the purposes of human actions and find three principal categories: achieving a goal or subgoal, collecting information to update the world model, and serving both purposes. This

distinction is consistent with WorldTest's design, which separates gathering information to learn a world model from using that model for downstream tasks (Warrier et al., 2026). Specifically, among human actions, 47.2% pursue goals, 9.1% collect information, and 28.4% serve both purposes. The corresponding proportions for successful LLM trajectories are 57.1%, 11.9%, and 30.5%.

For example, a TU93 participant proposes approaching and eating enemies, then follows the path and eliminates them, illustrating goal pursuit (Figure 5). In CN04, another participant attempts to connect shapes while saying they will "try and see what happens," working toward the connection goal while learning which configurations work (Figure 6).

Further analysis shows that exploration serves several purposes. Sometimes people explore an unfamiliar situation without an existing hypothesis to find out what happens. For example, an LF52 participant says they do not know what the newly introduced purple objects are, then clicks them and tries jumping a green object across them (Figure 7). Sometimes people act to test a proposed mechanism: an SC25 participant hypothesizes that a yellow pattern triggers teleportation, draws the pattern, and checks the resulting position (Figure 8). At other times, people investigate unexpected outcomes. Later in SC25, the same participant arrives somewhere other than expected, revisits earlier actions and tries other patterns, then proposes that the character's size may affect the result (Figure 9).

Among human actions with an information purpose, including those that also pursue a goal, unfamiliar-situation exploration, hypothesis testing, and investigation of unexpected outcomes account for 40.6%, 49.2%, and 9.8%, respectively; 0.1% combine these situations and 0.3% lack sufficient evidence for a subtype. For LLMs, the corresponding proportions are 10.7%, 49.0%, and 4.8%, with 35.5% combining situations. Hypothesis testing is common in both groups, while humans more often explore without a prior hypothesis and LLM information-seeking segments more often combine these situations.

4.4 Errors and recovery

We find that humans make errors while forming world models, planning, and executing actions, and can revise their understanding and actions through further interaction. Across 144 human trajectories, we record 15 accepted conclusions that conflict with the actual mechanisms, with subsequent corrections observed in 11 cases. For example, in game CN04, a participant initially believes that a shape must enter another shape before its length can change. They later discover that clicking its white endpoint is sufficient and explicitly retract the earlier explanation (Figure 10). We also record 79 occasions on which participants explicitly express that they do not understand a rule and offer no concrete explanation, and 57 occasions on which the level goal remains undetermined. In LF52, a participant repeatedly clicks without understanding how to move objects along the tracks, then discovers the arrow-key controls and recognizes that multiple platforms move together (Figure 11). Another excerpt from CN04 shows a participant progressing from uncertainty about how the red endpoints should meet to understanding how the shapes should fit together (Figure 12).

Human actions also exhibit planning failures and execution deviations. We record 1002 attempts that fail to achieve an intended goal or subgoal, with subsequent recovery observed in 916 cases. In LP85, a participant plans to position the orange object before moving the yellow one, but discovers that the latter operation also moves the orange object and then reconsiders their coupled movement (Figure 13). We further confirm 136 execution deviations in which the actual input differs from the intended operation, with 134 subsequently corrected. For example, a participant clicks outside a control's boundary and, after observing no effect, adjusts the click position and continues (Figure 14). These cases show how unsuccessful plans or input deviations lead to action adjustments and further progress.

The 119 successful LLM+harness trajectories also contain errors and recovery: 451 of 593 accepted but incorrect mechanism conclusions are subsequently corrected. We additionally record 29 occasions of explicitly not understanding a rule without a concrete explanation, and 83 occasions of an undetermined level goal. For LLMs, goal uncertainty often coexists with concrete candidates: 79 of the 83 occasions include one or more candidate goals, compared with 36 of 57 for humans. For example, Schema–Claude explicitly states that the goal in AR25 remains unknown and explores alternatives such as aligning a shape with a target or fitting it into a template, before using interaction feedback to identify the current level's winning condition. The larger number of recorded incorrect

mechanism conclusions for LLMs may partly reflect how their reasoning is expressed: public reasoning, code, and notes articulate mechanism understanding as explicit, checkable claims, making these conclusions and their revisions easier to observe. Human concurrent verbal reports may not express every change in understanding.

LLM trajectories also contain 822 attempts that fail to achieve an intended goal or subgoal, with subsequent recovery in 728 cases, and seven confirmed execution deviations, six of which are subsequently corrected.

5 Implications for Benchmark Design

Current LLMs, supported by effective harnesses, can perform well on ARC-AGI-3 tasks. By connecting action trajectories with concurrent thought reports, our dataset helps characterize how humans build, use, and revise world models in unfamiliar environments. These observations inform the design of next-generation benchmarks for interactive world-model learning.

Exploring counterfactual and emergent mechanisms. Both humans and LLMs infer some mechanisms before triggering them and use analogies to familiar concepts to understand the environment (Section 4.1). Such priors allow some mechanisms to be predicted without elaborate exploration. Future benchmarks could reduce reliance on familiar priors through counterfactual mechanisms. Mars (Tang et al., 2024), for example, changes terrain, character behavior, and task dependencies to depart from commonsense expectations, requiring learners to discover the actual rules through interaction. Another direction is emergent mechanisms: developers specify local dynamics rather than directly encoding complete higher-level mechanisms, whose effects arise from interactions among components. Learners would need to construct and test combinations to discover configurations that produce useful effects.

Induction through accumulated evidence and abstraction. Many current ARC-AGI-3 mechanisms can be understood from local, directly visible state changes, leaving room for more demanding inference. Among mechanism instances with both trigger and recognition annotations, the mean action difference between first recognition and first trigger is 90.8 for humans and 24.4 for LLMs with harnesses. Future benchmarks could draw on AutumnBench's hidden states and stochastic outcomes (Warrier et al., 2026), and introduce delayed effects to require evidence accumulated across interventions and inference about unobserved causal relationships. Humans also frequently guide actions with qualitative representations that omit precise details, more often than LLMs do (Section 4.2). The big world hypothesis emphasizes that an environment's complexity can far exceed an agent's finite memory and computation (Javed & Sutton, 2024). Benchmarks could accordingly make complete, precise pixel-level modeling infeasible, requiring qualitative abstractions that serve current goals and posing new challenges for LLM world-model learning.

Planning useful subgoals and testing combined outcomes. Future ARC-AGI-3 tasks could require longer plans and deeper subgoal dependencies. After recognizing all relevant main mechanisms, humans take 95.1 further actions on average to complete eligible levels, versus 49.2 for LLMs with harnesses. Constructing a device such as an automated sorting and transport system would pose two challenges. First, learners must derive useful subgoals from the intended function, identifying and building the required components. Second, even known local mechanisms may have uncertain combined effects, requiring learners to test the assembled device and revise their plans. Planning would thus combine subgoal construction with experimentation.

6 Conclusion

We introduce a dataset of human think-aloud reports aligned with gameplay on ARC-AGI-3, paired with LLM trajectories and annotations of world-model changes, action purposes, and errors and recovery. Our analyses highlight how learners use prior knowledge, reason with qualitative representations, explore, and revise their understanding and actions. These records support further research on interactive learning and motivate next-generation benchmarks that challenge familiar priors and require cumulative learning, abstraction, and planning.

ETHICS STATEMENT

This study was approved by an institutional review board (details withheld for anonymous review). All participants provided informed consent and were compensated for their time through Prolific.

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A HUMAN DATA COLLECTION PROTOCOL

The live-interaction interface introduces the study before presenting a controls tutorial and a separate practice game. Participants provide informed consent and enable their microphone; the microphone check asks them to say a sentence and detects speech before continuing. The study overview presents the following instructions:

Study instructions

You will play 3 different video games. The goal of each game is unknown. You need to experiment, infer what the goal might be, and speak your thought process into the microphone at the same time.
Try to win each game while thinking aloud continuously. Say everything that is going through your mind as you play.
We care primarily about how detailed your think-aloud explanation is—not how quickly you finish. We estimate about 20 minutes per game, but it is fine to take longer; you will be compensated for the additional time.

The controls tutorial explains that available controls vary by game and unavailable controls are disabled. It introduces directional input using on-screen buttons or arrow keys, the space action, clicking directly on objects or positions, undo when supported, and resetting the current level. It notes that the practice game does not support the space or undo actions. These instructions explain interface operations without supplying the mechanisms or solutions of the formal games. Before practice, participants are told:

Practice instructions

You will now have 10 minutes to beat one practice game. Keep thinking aloud from start to finish—say everything that is going through your mind. Your microphone will record continuously.

Practice has a 10-minute limit, and completing the practice game unlocks the formal study. Giving up or reaching the practice deadline without completing the game ends participation. The formal study assigns three games by default, each with a 60-minute maximum, and includes the games' full level sequences. The 20-minute duration in the overview is an estimate communicated to participants, not the enforced time limit. Giving up a formal game or reaching its time limit ends the session without offering the remaining games.

During both practice and formal play, approximately 10 seconds without detected speech triggers a reminder to continue verbalizing current thoughts. The reminder disappears when speech resumes and provides no game-specific advice. The timer excludes time spent in a background browser tab, and reminders are suppressed during instructional overlays and terminal states. The instructions communicate compensation for additional time; the exit message for Prolific participants explains that compensation for submitted participation is paid through a Prolific bonus after returning the study.

The reported 144 trajectories constitute the human analysis corpus across the 25 public games, rather than all assigned attempts or practice runs. As described in Section 3.2, the human analysis uses completed games. The default assignment of three games therefore does not imply three retained trajectories per participant.

B ANNOTATION PROCEDURES AND REVIEW DETAILS

B.1 MECHANISM CATALOG

Using the definition in Section 3.1, we catalog mechanisms that can independently change a player's prediction, action choice, or risk assessment. For example, clicking a green support removes it in Table 1. For every game and level, two authors inspect the game source, rendered states, and representative replays and jointly catalog the key mechanisms through discussion. We split an engine

event when it produces multiple independently meaningful effects, and exclude rendering details, cosmetic changes, one-off solution steps, and the mere reuse of an earlier mechanism in a harder layout. Each mechanism receives a stable identifier, a natural-language definition, its earliest observable level, and a scope label. The primary analysis uses *main* mechanisms; optional failure, undo, or resource-exhaustion behaviors are retained as supplemental mechanisms. We denote the mechanism set for game g by $\mathcal{M}_g$.

B.2 TRIGGER DETECTION

We insert mechanism-specific trigger-detection code into the ARC game source and execute the instrumented game during trajectory replay. The detection code checks changes in selected internal variables before and after each action, as well as whether relevant code paths execute. We use an LLM as a code-generation assistant to turn each mechanism definition into a candidate runtime monitor. The LLM is given the relevant game source, the mechanism description, and representative before/after states. It proposes checks over state changes, actions, and distinctive source paths. Annotators review the proposed code, run it against positive and negative replays, and revise it until the event has a precise player-level meaning. A validated monitor has the form

$$d_m(s_{a-1}, u_a, s_a, c_a) \in \{0, 1\}, \qquad t^x_{\tau m} = \min\{a : d_m = 1\}, \tag{2}$$

where u_a is the ath player action, s_{a-1} and s_a are the game states before and after that action, including the monitored internal variables, c_a records whether the relevant code paths executed, and $t^x_{\tau m}$ is the first action at which mechanism m is objectively triggered in trajectory τ. The LLM proposes monitor code; the accepted mechanism definition, binding, and test result are human verified. Monitor activation records objective execution of a known mechanism and does not label whether the player understood it. We manually inspected 30 trigger annotations and found all 30 to be correct.

B.3 RECOGNITION EVIDENCE AND REVIEW COVERAGE

Recognition timing is annotated from the full chronological trajectory context. Agent and human records are annotated separately with LLM assistance to preserve their different evidence modalities. Agent inputs contain the mechanism definition, public outputs and any released reasoning, harness messages, aligned actions, and visible outcomes. Human inputs contain the same definition, aligned think-aloud transcript, actions, and outcomes. For each trajectory–mechanism pair, $t^r_{\tau m}$ denotes the first action with sufficient evidence of recognizing mechanism m. The annotation also records supporting text or behavior, earlier partial or incorrect beliefs, and later corrections or drift.

We accept two kinds of recognition evidence. *Explicit evidence* is a public statement that correctly specifies the relevant causal mechanism, condition, direction, object, or goal. *Strong behavioral evidence* is a purposeful prediction, diagnostic action, or plan followed by an outcome that rules out live alternative explanations. Accidental triggering, ordinary movement, a forced route, or level completion alone is insufficient. The annotation distinguishes recognized, partial-only, available-but-not- recognized, not-yet-observable, and unreviewable cases.

Recognition review checks the quoted evidence against the original log and action context, using the frozen game source and relevant replay states to assess whether the stated mechanism is correct. When a human verbal reference or visual consequence is ambiguous, we consult the aligned replay to interpret the report in its action context. LLM annotators review human and agent records chronologically, checking the available verbal or reasoning evidence against the action histories, game source, and replay states. Labels are not assigned from keyword matches or successful completion alone. We manually checked 100 annotations, of which 97 were correct. Each decision identifies exact evidence, its action boundary, and later retention or drift. Source code establishes the environmental mechanism, not whether the acting model knew it. The default analysis uses the adjudicated best-evidence timestamp; explicit-only and behavior-only timestamps are retained for sensitivity checks where separately supported.

B.4 ACTION PURPOSES, ERRORS, AND RECOVERY

Following Section 3.3.2, we classify contiguous human and LLM action segments as goal pursuit, information gathering, or both. Information gathering includes exploring unfamiliar situations, test-

ing hypotheses, and investigating unexpected outcomes. Labels use the aligned verbal or reasoning evidence and surrounding observations. GPT-5.6 Sol assists classification; one author reviewed 30 segments and found all 30 correct.

Error annotations distinguish accepted but incorrect mechanism beliefs, attempts that fail to achieve an explicit goal or subgoal, and inputs that deviate from the intended operation (Section 3.3.3). Tentative hypotheses, explicitly not understanding a rule, and uncertainty about the level goal are distinguished from accepted incorrect conclusions. Continuous expressions about the same issue are merged into one event.

Each error is linked to later evidence that corrects the same belief, plan, or input; cases without observed correction are retained. Annotations preserve supporting statements, actions, and outcomes, with the game source and replay establishing the relevant mechanism. Section 4.4 reports event counts and observed corrections.

C HUMAN TRAJECTORIES AND CONCURRENT VERBAL REPORTS

These excerpts show humans drawing on prior knowledge, pursuing goals, gathering information, revising their understanding, and adjusting their actions. Game frames and original verbal reports are aligned independently to the action axis; pale bands and connectors locate the estimated speech intervals. A checkmark indicates level completion in place of the automatic next-level observation. Horizontal spacing is schematic, and ellipses indicate omitted actions.

C.1 PRIOR KNOWLEDGE

These excerpts illustrate the two human cases in Section 4.1.

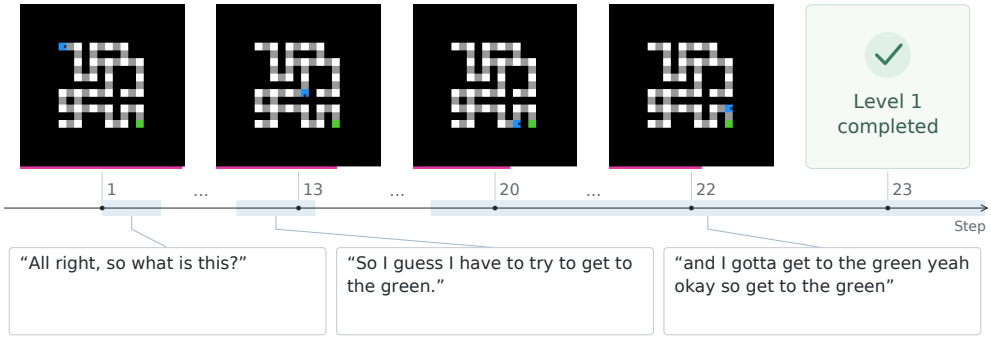


Figure 3: **Inferring a goal before reaching it (TU93).** The participant identifies the green location as a candidate goal before first reaching it.

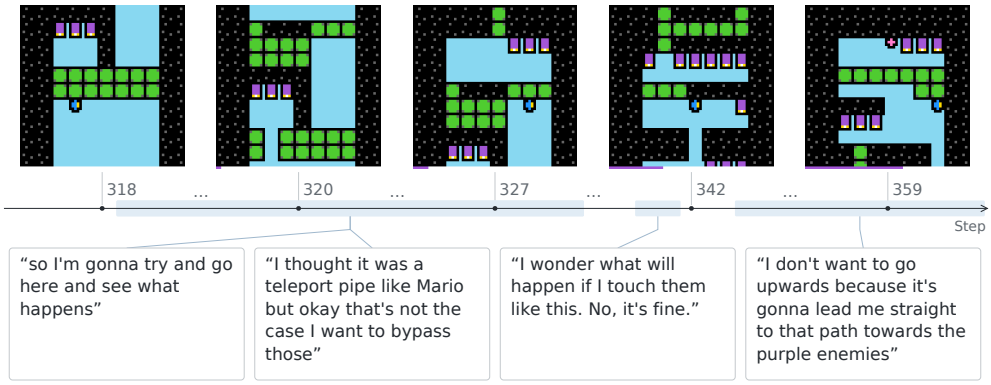


Figure 4: **Revising an analogy-based hypothesis (BP35).** The participant retrospectively describes a Mario-style teleport-pipe interpretation, rejects it, and tries another route.

C.2 ACTION PURPOSES

These excerpts illustrate goal pursuit, information gathering, and their combination in Section 4.3. The information-gathering examples show exploration of unfamiliar objects, hypothesis testing, and investigation prompted by an unexpected outcome.



Figure 5: **Pursuing a subgoal (TU93).** The participant plans an approach and executes a sequence that removes the enemies.



Figure 6: **Combining goal pursuit and information gathering (CN04).** Testing candidate shape connections both advances the solution and reveals which configurations work.

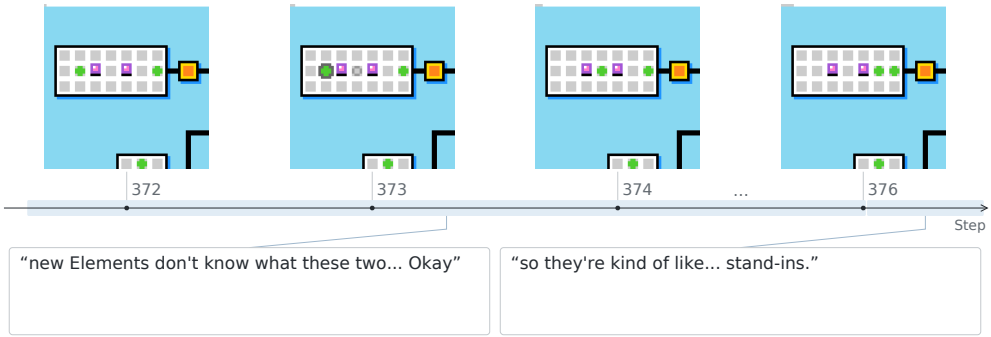


Figure 7: **Exploring unfamiliar objects (LF52).** The participant tests clicks and movement before proposing an explanation for the purple objects.

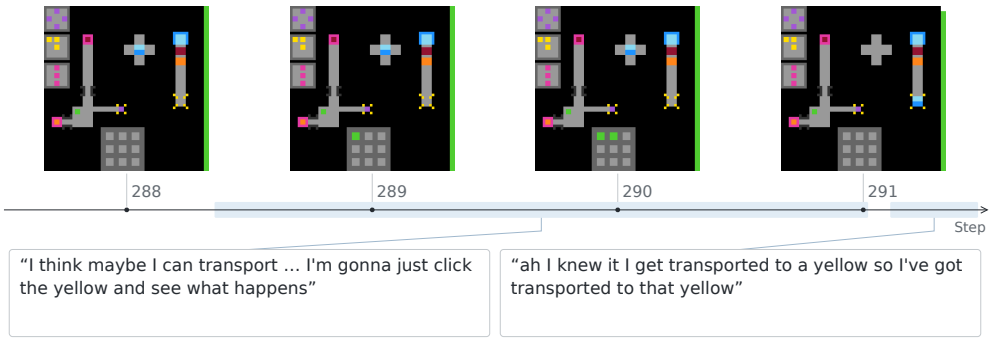


Figure 8: **Testing a hypothesis (SC25).** The participant predicts teleportation, tests the yellow pattern, and confirms the predicted effect.

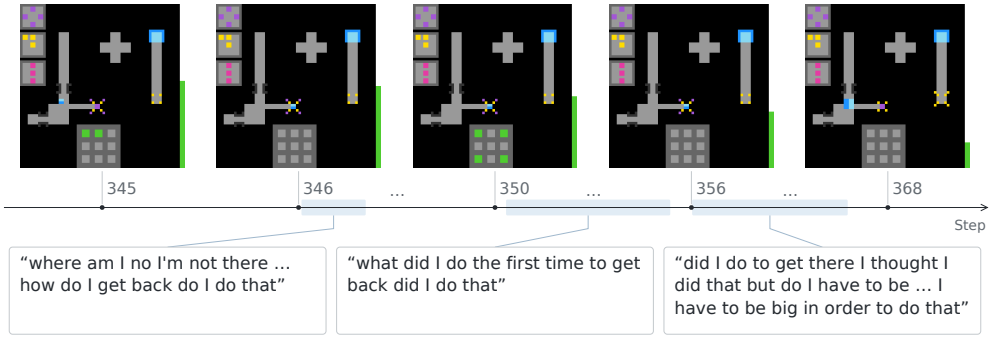


Figure 9: **Investigating a prediction error (SC25).** An unexpected teleport destination prompts further tests and the hypothesis that returning requires a larger character.

C.3 Revised understanding and action

The following excerpts illustrate the human cases discussed in Section 4.4.

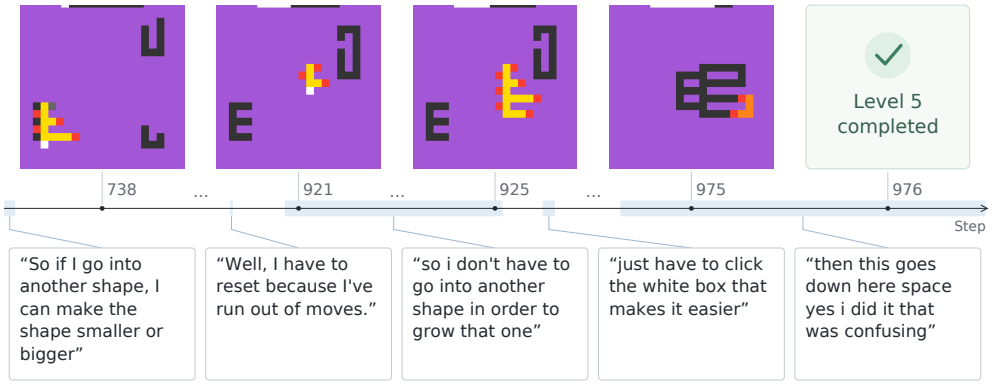


Figure 10: **Revising a mechanism condition (CN04).** Resizing in place leads the participant to retract the belief that entering another shape is necessary.

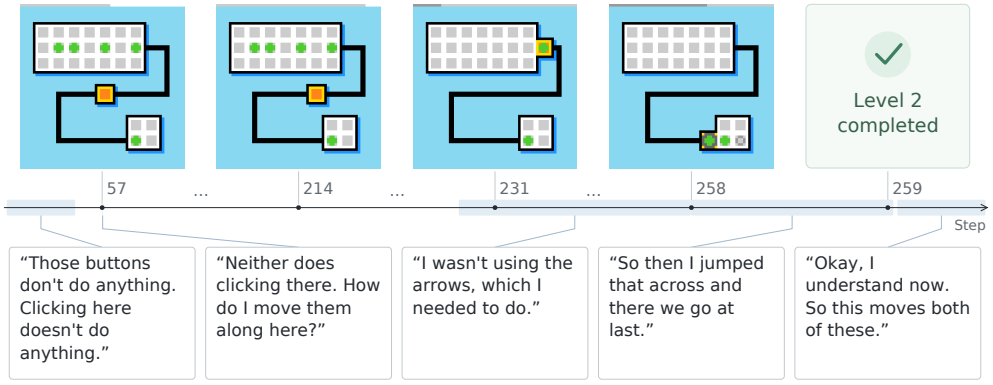


Figure 11: **Discovering the controls (LF52).** After unsuccessful clicks, the participant identifies the arrow-key controls and completes the level.

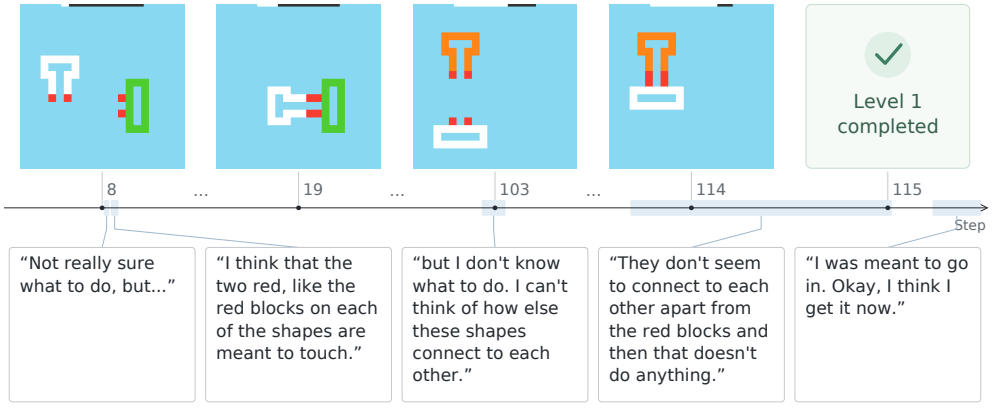


Figure 12: **Clarifying the connection goal (CN04).** The participant tests how the shapes should fit together and reports understanding after completing the level.



Figure 13: **Replanning under coupled movement (LP85).** Moving the yellow object disrupts the orange object’s placement, prompting a revised sequence that places both at their targets.



Figure 14: **Correcting an input (LP85).** A click just outside the button has no effect; a small adjustment activates it.

D ADDITIONAL EXPERIMENTAL DETAILS AND RESULTS

D.1 SYSTEMS AND COMPARISON SCOPE

The 394 trajectories contain 236,806 player actions. All conditions use the same catalog of 354 mechanisms (320 main and 34 supplemental), yielding 5,565 trajectory–mechanism annotation instances.

Direct-agent configurations are Claude Opus 5 high, GPT-5.6 Sol max and high, and GPT-6 Astra max in the official Standard configuration. VISTA uses Claude Opus 5.0 or GPT-5.6 Sol max. Schema–Claude includes 14 Opus 4.8 and 11 Fable 5 runs, all at max effort; Schema–GPT includes 16 GPT-5.6 Sol xhigh and nine max runs. OPINE-World uses Claude Opus 4.8 high. Astra Standard and Provider Adapter use the same released model at max effort. Standard uses the benchmark's direct-agent interface with carried notes, whereas Adapter preserves additional reasoning state and supports conversation compaction. These system labels retain the configurations in the public releases.

Human and VISTA trajectories are selected for successful game completion; other conditions retain unsuccessful trajectories from the chosen releases. Model configuration, inference budgets, and the visibility of reasoning records are not controlled across systems. Comparisons are therefore descriptive rather than causal estimates of adding a harness. An absence of public recognition evidence does not establish an absence of internal mechanism knowledge. The six unsuccessful harness traces consist of one Schema–GPT and five OPINE–Claude trajectories; they do not estimate a general harness failure rate.

D.2 COVERAGE AND CONDITIONAL ACTION DISTANCES

Tables 2–4 provide the coverage and denominators for the recognition-timing comparisons in Section 4.1 and action distances discussed in Section 5. Observable instances are main mechanisms introduced at levels the trajectory has reached. The first action within a level has exploration distance one. Means and medians use uncapped action coordinates. A mechanism without a trigger or supported recognition is excluded from the corresponding distance rather than assigned zero. These measures describe overlapping aspects of interaction, not mutually exclusive cognitive phases.

Table 2: Trigger coverage and supplementary exploration statistics. Observable instances are main mechanisms introduced at reached levels.

Condition	Triggered/obs.	Coverage	$D^{\text{exp}} \leq 5$
Claude direct	218/238	91.6%	42.2%
GPT-max direct	177/195	90.8%	46.9%
GPT-high direct	159/182	87.4%	39.6%
Astra Standard	266/285	93.3%	47.7%
VISTA–Claude	294/320	91.9%	54.1%
VISTA–GPT	305/320	95.3%	49.5%
Schema–Claude	301/320	94.1%	45.8%
Schema–GPT	296/320	92.5%	51.7%
OPINE–Claude	285/302	94.4%	42.5%
Astra Adapter	287/320	89.7%	55.7%
Human	1795/1825	98.4%	34.1%

Humans trigger 1,795 of 1,825 observable main-mechanism instances (98.4%); agent coverage ranges from 87.4% to 95.3%. Because the human records are completion-selected, this is not an exploration success rate for all recruited participants. Conditional distances can favor a system that reaches easy mechanisms but misses difficult ones, so timing and coverage answer different questions.

Human induction distances use 1,425 instances with both trigger and recognition annotations. Recognition coverage ranges from 63.7% to 84.0% for direct agents and from 41.2% to 93.1% for harness systems, compared with 78.1% for humans. Astra Adapter has supported recognition for 41.2% of observable mechanisms, versus 77.5% for Standard, despite completing 25 games versus Standard's 12. This illustrates why observed recognition coverage should not be equated with task

Table 3: Recognition coverage, paired instances, and recognition preceding a trigger.

Condition	Recognition rate	Paired/obs.	$D^{\text{ind}} < 0$
Claude direct	84.0%	193/238	11.4%
GPT-max direct	79.0%	150/195	12.7%
GPT-high direct	63.7%	114/182	13.2%
Astra Standard	77.5%	214/285	9.3%
VISTA–Claude	92.8%	284/320	11.3%
VISTA–GPT	93.1%	291/320	9.6%
Schema–Claude	89.4%	278/320	10.8%
Schema–GPT	87.8%	271/320	9.2%
OPINE–Claude	83.8%	247/302	7.3%
Astra Adapter	41.2%	126/320	5.6%
Human	78.1%	1425/1825	4.0%

competence: successful behavior may coexist with sparse mechanism-specific evidence in public records.

Recognition precedes the first monitored trigger in 5.6–13.2% of paired agent instances and 4.0% of human instances. Such cases can reflect inference from the initial scene, related events, or knowledge carried across levels; they do not by themselves establish memorization or pretraining exposure. Trigger definitions are mechanism-specific, so these signed action distances are not uniform reaction latencies. Evidence before and after an action shares its integer coordinate, with phase recorded separately.

Later drift is recorded in 55/221 recognized main-mechanism instances for Astra Standard, 4/132 for Adapter, 10/286 and 11/281 for the two Schema conditions, and 11/253 for OPINE. Both direct and harness-assisted models can thus abandon a previously supported mechanism. Differences in public evidence limit comparisons of drift frequency across systems.

Table 4: Eligible planning levels and missing recognition evidence.

Condition	Eligible/completed	Missing recognition
Claude direct	45/95	50
GPT-max direct	21/48	27
GPT-high direct	10/30	20
Astra Standard	14/137	123
VISTA–Claude	70/183	113
VISTA–GPT	79/183	104
Schema–Claude	61/183	122
Schema–GPT	59/182	123
OPINE–Claude	25/160	135
Astra Adapter	1/183	182
Human	228/1057	829

Planning distances use completed levels for which every main mechanism introduced by that level has recognition evidence by completion. Of 1,057 completed human levels, 228 qualify. Astra Standard has 14 qualifying levels out of 137 completed, and Adapter has one out of 183. Direct GPT-high's mean of 7.7 actions is based on ten completed levels, despite no full-game wins. These distances cannot characterize planning on levels without sufficient recognition evidence. Evidence first recorded after entry into the next level cannot qualify the preceding level, even when both share an action boundary. The interval does not isolate all planning activity or reconstruct later belief drift.